A Note on Averaging Day-Ahead Electricity Price Forecasts Across Calibration Windows

Initial Report: replication and extension

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1. Introduction

Electricity prices are hard to forecast. They change strongly during the day, they depend on the day of the week and on the season, and they sometimes jump when demand is close to the available capacity. For this reason day-ahead electricity price forecasting (EPF) is a research field of its own (Weron, 2014). A common modelling choice is the so-called multivariate framework, where 24 separate sub-models are estimated, one for each hour of the day (Ziel and Weron, 2018).

When these models are estimated, one has to decide how much past data to use, i.e. the length of the calibration window. A long window gives more precise parameter estimates, but the model reacts slowly when the market changes; a short window adapts quickly, but the estimates are noisier. The article we chose, Hubicka, Marcjasz and Weron (2019, DOI 10.1109/TSTE.2018.2869557), addresses exactly this problem. Instead of looking for one "optimal" window, the authors average the forecasts obtained from windows of different lengths, and show that this simple idea improves accuracy, even compared with the best single window chosen after seeing the results.

In this project we want to replicate the main ARX results of that article on the GEFCom2014 data (Hong et al., 2016) and then add a small extension. Section 2 summarises the article and Section 3 describes our plan.

2. Summary of the original article

Data and test period. The authors use the public GEFCom2014 price track, with hourly prices and hourly day-ahead forecasts of system load from 1 January 2011 to 17 December 2013. The largest window (728 days) sets the start of the out-of-sample period: the test sample runs from 29 December 2012 to 17 December 2013, i.e. 354 days. Forecasts are made with a rolling scheme, so the windows move forward by one day at a time.

Models. Two models are used, each made of 24 hourly sub-models. The first is an ARX model with log-prices $p_{d,h} = \log P_{d,h}$:

$$p_{d,h} = \beta_{h,0} + \beta_{h,1}\, p_{d-1,h} + \beta_{h,2}\, p_{d-2,h} + \beta_{h,3}\, p_{d-7,h} + \beta_{h,4}\, p_{\min} + \beta_{h,5}\, z_t + \Sigma_{i\in\{1,6,7\}}\, \beta_{h,i+5}\, D_i + \varepsilon_{d,h}$$

where $p_{\min}$ is the minimum of the 24 prices of the previous day (it links all hours of yesterday to today's price), z_t is the log of the day-ahead load forecast, and D_i are dummies for Monday (i = 1), Saturday (i = 6) and Sunday (i = 7). The second model is a NARX neural network with the same inputs, one hidden layer with 5 neurons and a linear output. Because network training starts from a random point, each forecast is the average of 5 training runs (a committee machine; Marcjasz, Uniejewski and Weron, 2019).

Windows and averaging. The models are estimated on windows of T = 28, 56, …, 728 days (multiples of 7, to respect the weekly seasonality). The forecast from one window is denoted Win(T), and the average of the forecasts from a set of windows is denoted AW(T), using Matlab notation: for example AW(28:28:728) averages 26 windows. Accuracy is measured with the Mean Absolute Error (MAE) over all test days and all 24 hours.

Main results. (i) The average over all windows, AW(28:728), has a lower MAE than every single window Win(T), for both models, and it is about 1.2–1.35% better than the longest window Win(728) (relative change computed as the log of the MAE ratio). (ii) Averaging over a few well-chosen windows is better still and much cheaper. Starting from {28, 728} and adding windows in pairs (one longer short window and one shorter long window), the lowest MAEs are reached with as few as 4 windows for ARX and 6 for NARX. The authors recommend AW(28:28:84,714:7:728), i.e. six windows of 28, 56, 84, 714, 721 and 728 days,

which works well for both models. The gains are larger for ARX than for NARX. (iii) To check significance, they use the multivariate Diebold–Mariano test (Uniejewski, Weron and Ziel, 2018), where the loss of a model on one day is the sum of its absolute errors over the 24 hours. For NARX, the recommended set is significantly better than all the other sets at the 5% level; for ARX, the best four sets are not significantly different from each other, so the recommended set is not significantly outperformed by any other. The conclusion mentions more advanced models (SVMs, LASSO-estimated regressions) and more sophisticated weighting schemes as future work.

3. Proposed extension

We tried to keep the extension realistic, because time is limited and we do not want to promise what we cannot deliver. Both parts reuse the forecasts of the individual windows, so the second part needs little extra computation.

Part 1: other ways of averaging. The article combines windows with the simple arithmetic mean. We want to check whether other rules work better, since electricity prices have occasional large jumps. All rules will be compared with the arithmetic mean using the same MAE, so the only difference is the aggregation rule:

(a) the median of the window forecasts, for the six-window set AW(28:28:84,714:7:728) and for the full set AW(28:28:728);

(b) a 10% trimmed mean (10% removed from each tail), for the full set of 26 windows only, because with six forecasts a 10% trim would remove nothing.

Part 2: convex combination with inverse-MAE weights. Following a suggestion from our professor, we will combine the forecasts of ten windows $i = 1, \ldots, 10$ with weights $w_i = \mathrm{MAE}_i^{-1} / \Sigma_j \mathrm{MAE}_j^{-1}$, where the sum is over the ten windows. Then $w_i > 0$ and the weights sum to 1, so the combined forecast is a convex combination: windows that were more accurate recently get a larger weight. MAE_i is computed only on the N days before the forecast day (N = 60 as baseline, N = 7 and 28 as checks), so no future information is used; for the first test days we use equal weights. The ten windows will be a mix of short and long ones, as in the article (for example 28:28:140 and 700:7:728). We compare the result with the arithmetic mean of the same windows and with AW(28:28:84,714:7:728), using the MAE and the Diebold–Mariano test. A weighting based on past performance was already proposed by Marcjasz, Serafin and Weron (2018), so we see this as a check of that idea on the GEFCom2014 data and not as a new method.

What we will not do. We will not implement the NARX model. Doing it for all windows would need 24 hours × 5 committee runs × about 353 test days × 26 windows, which is more than one million network trainings, and this is not feasible for us. We will report and discuss the NARX results of the article, but we will not reproduce them.

Status so far. We use the GEFCom2014 price file for zone 1. So far we have implemented the loading of the data, the day × hour matrices of log-prices and log-load forecasts, the removal of the days with missing prices, and the functions that build the rolling training sample and compute the one-day-ahead ARX forecast for a given hour. We still have to run the 24 hourly regressions over all test days and windows, compute the averages AW and the MAE, and run the Diebold–Mariano test. In our file the prices of the last hours of 17 December 2013 are missing, so our test period ends on 16 December 2013 and has 353 days instead of 354; we will state this in the final report.

Timeline. We plan to finish the replication of the ARX results and the Diebold–Mariano test by the end of November, in time for the progress report on 1–2 December. In December we will implement the new averaging rules and the inverse-MAE combination. The final report will be submitted by 8 January 2027.

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